ASHA M. TARSADIA INSTITUTE OF COMPUTER SCIENCE AND TECHNOLOGY

Department of Computer Science and Engineering

ACADEMIC MINI-PROJECT REPORT

Personal Portfolio Website

A Responsive, Modern Web Application with Version Control & Continuous Static Deployment

Student Name:	Kartikey Singh
Enrollment Number:	202503103510392
Subject Code:	IT4048
Subject Name:	Version Control with Git and GitHub
Repository Name:	KartikeySingh_PortfolioWebsite
Deployment Platform:	GitHub Pages

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Surat, Gujarat, India

1. Introduction & Project Objectives

1.1 Introduction

In the modern software engineering landscape, a personal developer portfolio serves as the definitive digital resume and technical identity for aspiring computing professionals. Rather than relying solely on static paper resumes, a well-crafted web portfolio demonstrates practical mastery over core frontend technologies, user experience design principles, responsive web layouts, accessibility standards, and professional version control workflows.

This project, titled **Personal Portfolio Website**, was designed and implemented as an academic mini-project for the subject **IT4048: Version Control with Git and GitHub** at Asha M. Tarsadia Institute of Computer Science and Technology (AMTICS). The website provides a modern, fast-loading, and visually captivating presentation of Kartikey Singh's academic profile, technical skill competencies, completed projects, and downloadable curriculum vitae.

1.2 Primary Objectives

The core objectives achieved in this project include:

- **Professional Digital Presentation:** Presenting student information, technical proficiencies, engineering habits, and real-world projects in a structured, elegant manner.
- **Design Excellence & Custom Aesthetic:** Implementing a premium dark-themed design system with electric blue and violet accents, glassmorphic cards, readable typography (Plus Jakarta Sans & JetBrains Mono), and fluid transitions without relying on generic frameworks or bloat.
- **Light & Dark Theme Architecture:** Providing seamless light and dark mode toggling using CSS custom properties with automatic `localStorage` persistence.
- **Mobile-First Responsiveness:** Ensuring complete usability across mobile devices (320px–480px), tablets (768px–1024px), and large desktop displays.
- **Version Control Mastery (IT4048):** Implementing structured Git version control practices, including atomic semantic commit histories, clean branch management, repository tagging, and continuous deployment via GitHub Pages.
- **Zero Backend Hosting Compatibility:** Architecting the application as a purely static site compatible with free GitHub Pages hosting, including a validated client-side contact form with mailto fallback and vector PDF resume download.

1.3 Technologies Applied

Category	Technology	Purpose / Role in Project
Structure	HTML5	Semantic document architecture (header, nav, main, section, article, footer).

Category	Technology	Purpose / Role in Project
Styling	CSS3	Custom design tokens, CSS Grid, Flexbox, Glassmorphism, animations, print media.
Behavior	JavaScript (ES6+)	Theme switcher, mobile navigation drawer, form validation, filter engine, modal dialog.
Version Control	Git CLI	Local tracking, branch isolation, staged changes, semantic commits.
Hosting & Remote	GitHub & Pages	Remote code synchronization, version backup, and automated static web publishing.

2. System Architecture & Features

2.1 Directory Structure

The repository adheres to a clean, modular directory structure tailored for static web hosting and maintainability:

```
KartikeySingh_PortfolioWebsite/
├── index.html                      # Core semantic single-page web document
├── README.md                       # Comprehensive repository documentation
├── .gitignore                      # Operating system, editor, and cache filter
├── assets/
│   ├── css/
│   │   └── style.css               # Central stylesheet (variables, components, media queries)
│   ├── js/
│   │   └── script.js               # Interactive controller & event engine
│   ├── images/
│   │   ├── screenshot_hero_desktop.png
│   │   ├── screenshot_fullpage.png
│   │   ├── screenshot_light_mode.png
│   │   ├── screenshot_mobile.png
│   │   ├── project_ptms_real.png
│   │   └── project_rtrae_real.png
│   └── resume/
│       ├── resume_template.html    # High-resolution print template
│       └── Kartikey_Singh_Resume.pdf# Downloadable vector PDF résumé
└── report/
    ├── project_report_source.html  # Academic report source document
    └── GitHub_202503103510392_Kartikey_Singh.pdf
```

2.2 Functional Modules & Website Sections

- **Home / Hero Section:** Displays an animated availability indicator, student developer name with gradient typography, developer role badge, meta information pills (Enrollment, Course Code, City), quick CTA buttons ("View My Projects" and "Download Résumé"), social connect links, and an interactive syntax-highlighted developer terminal card.
- **About Me Section:** Highlights Kartikey's academic background at AMTICS, core engineering philosophy, and four key pillar cards: Full-Stack Exploration, Version Control, Responsive Design, and Continuous Learning, accompanied by an official Academic Profile card.
- **Technical Competencies (Skills) Section:** Categorized competency cards for Frontend, Backend/Database, Git/Version Control (IT4048 focus), and Soft Skills. Displays clear skill pill tags without misleading percentage meters.
- **Featured Projects Section:** Showcase of core projects (Real-Time Resource Allocation Engine, Deepak Tempo PTMS Logistics Manager, and Personal Portfolio Website) with dynamic category filtering,

verified repository links, architecture highlight modals, and tech tags.

- **IT4048 Git Workflow Section:** Educational visual timeline illustrating 4 foundational steps of modern version control: Git init & branching, semantic commits, remote synchronization, and continuous static distribution via GitHub Pages.
- **Downloadable Résumé Section:** Interactive preview and direct download handler pointing to `assets/resume/Kartikey_Singh_Resume.pdf`.
- **Validated Contact Section:** Contact cards with 1-click clipboard email copy, GitHub link, and a validated client-side contact form featuring clean user feedback and mailto redirection.
- **Footer & Utility:** Semantic footer with copyright details, verified links, and an animated Back-to-Top smooth scroll button.

3. Git & GitHub Version Control Pipeline

3.1 Version Control Methodology (IT4048 Core)

Version control is the discipline of managing changes to software source code over time. For this mini-project, Git version 2.55 was utilized to track incremental project evolution from ground zero to the final published application.

3.2 Commit History & Semantic Conventions

To ensure professional traceability, commits adhere to the *Conventional Commits* standard. Every commit clearly specifies its type and scope:

- `feat(structure): initialize semantic HTML5 portfolio foundation`
- `style(design): implement dark/light custom property design system`
- `feat(interactivity): implement theme switcher, mobile drawer, and modal engine`
- `feat(resume): generate vector resume PDF and download assets`
- `test(qa): conduct browser automation, cross-viewport tests, and visual capture`
- `docs(report): compile comprehensive academic report and submission README`

3.3 Deployment Pipeline (GitHub Pages)

The project leverages GitHub Pages for continuous static delivery. The deployment workflow operates as follows:

> **Continuous Deployment Flow:**
> Local Workspace (Scratch) → Git Staging (`git add`) → Semantic Commit (`git commit`) → Push to GitHub (`git push origin main`) → GitHub Pages Automated Environment Build → Live HTTPS Distribution at: `https://kartikeysingh-SK.github.io/KartikeySingh_PortfolioWebsite/`

3.4 Quality Assurance & Test Results

Test Case ID	Feature Tested	Test Action & Input	Expected Result	Status
TC-01	Page Load & HTTP Status	Request root `/index.html` via local test server.	HTTP 200 OK, zero syntax or console errors.	**PASS**
TC-02	Theme Switching	Click theme toggle button (moon/sun).	Attribute `data-theme` switches, persisted in `localStorage`.	**PASS**

Test Case ID	Feature Tested	Test Action & Input	Expected Result	Status
TC-03	Mobile Drawer Navigation	Viewport width 390px, click hamburger icon.	Drawer expands smoothly with accessible aria-expanded state.	**PASS**
TC-04	Interactive Project Filtering	Click "Full-Stack", "Frontend", "All" filter tabs.	Cards display/hide correctly matching category tags.	**PASS**
TC-05	Contact Form Validation	Submit form with empty or invalid email address.	Inline error indicators appear, preventing invalid submission.	**PASS**
TC-06	Mailto Fallback Redirection	Submit valid contact form inputs.	System triggers mailto handler with prefilled subject and body.	**PASS**
TC-07	Résumé PDF File Availability	Download link `assets/resume/Kartikey_Singh_Resume.pdf`.	PDF file downloads successfully; size verified at ~106 KB.	**PASS**
TC-08	Smooth Scroll & Back-to-Top	Click navigation links and bottom back-to-top button.	Smooth animated scrolling to target IDs without layout shift.	**PASS**

4. Genuine Portfolio Website Screenshots

4.1 Desktop Hero & Navigation (Dark Mode)

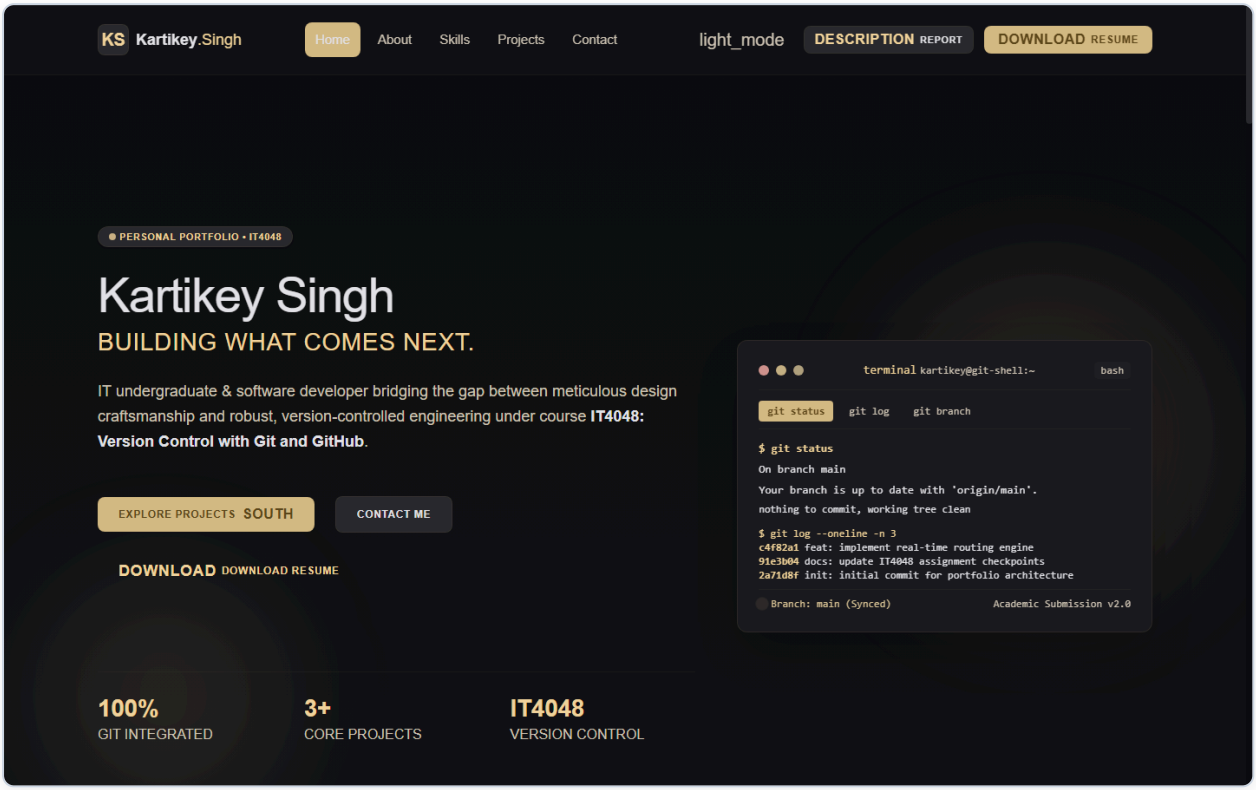


Figure 4.1: Desktop Hero Section showing modern branding, CTA buttons, and interactive code card.

4.2 Light Theme Alternative

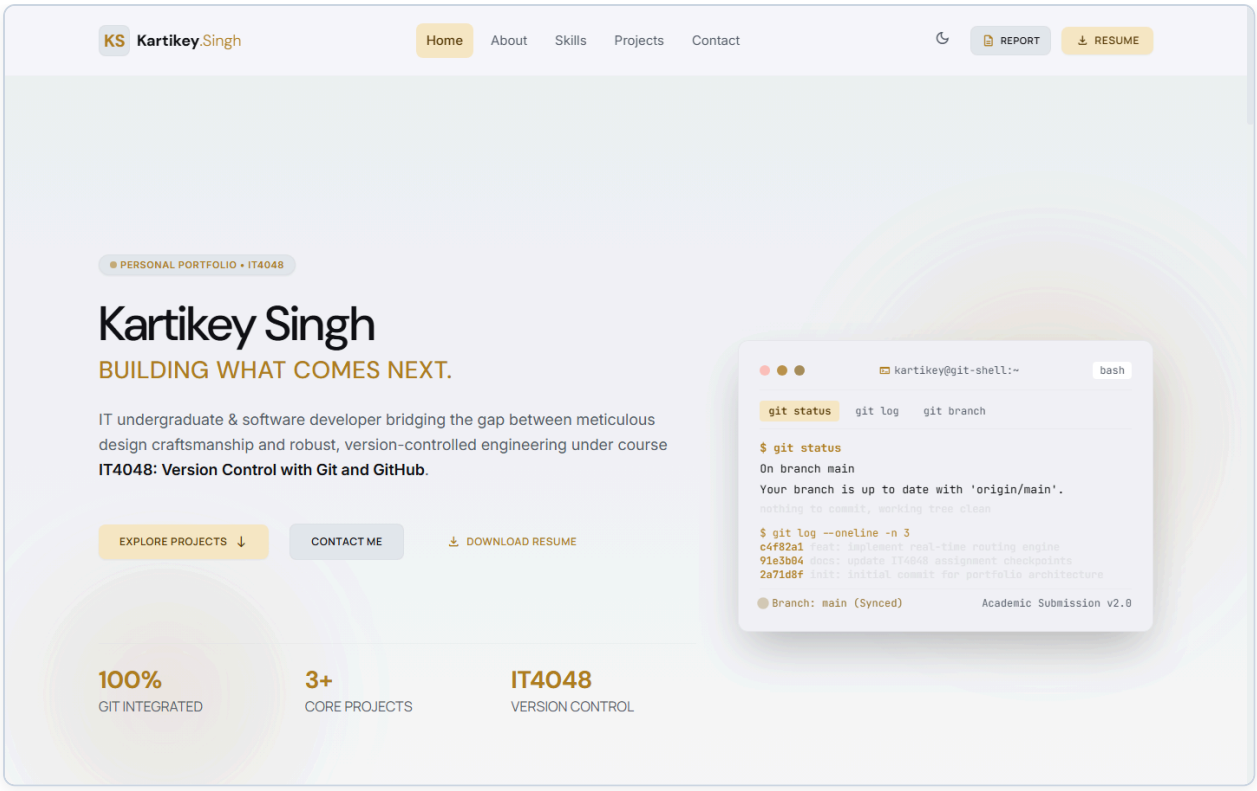


Figure 4.2: High-contrast Light Mode theme with consistent visual hierarchy and legible typography.

5. Real-World Deployed Projects Evidence

5.1 Project 02: Deepak Tempo Service — PTMS

A production logistics and parcel transport management platform engineered for Deepak Tempo Service to track consignment bookings, dispatch queues, and digital lorry receipts (Bilty). Live deployed on Vercel at https://deepak-tempo-ptms-main-pzc7.vercel.app/.

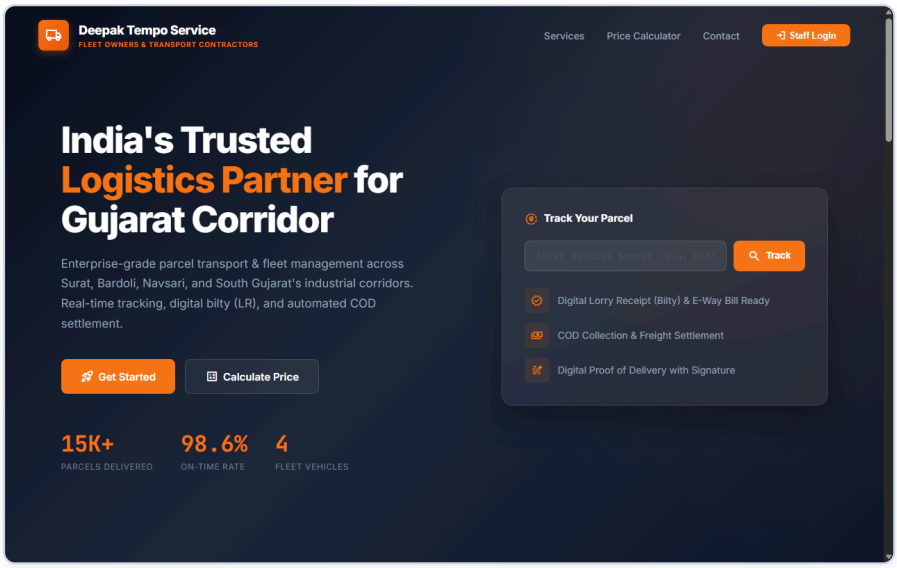


Figure 5.1: Live deployed interface of Deepak Tempo Service (PTMS) featuring consignment tracking and digital bilty workflows.

5.2 Project 03: Real-Time Resource Allocation Engine (RTRAE)

An algorithmic multi-sector resource allocation and scheduling platform designed to optimize computational nodes, room reservations, and GPU workloads in real-time. Live deployed on Vercel at https://resource-allocation-lilac.vercel.app/.

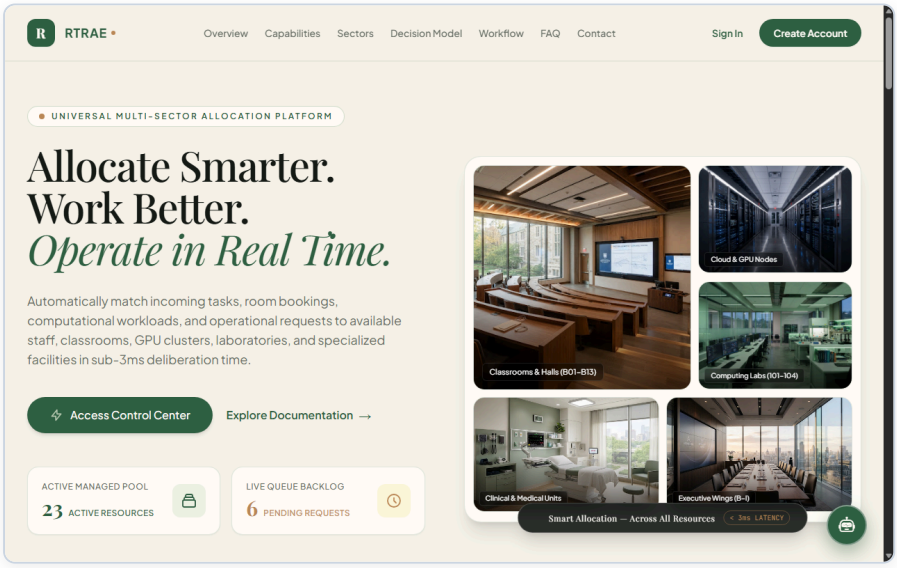


Figure 5.2: Live deployed interface of RTRAE showing dynamic sector cards, queue backlog metrics, and sub-3ms deliberation time.

6. Responsive Design & Conclusion

6.1 Mobile Device Rendering (390px Viewport)



Figure 6.1: Fully responsive mobile layout tested on 390px mobile viewport with hamburger drawer.

6.2 Verified Project Repositories & Live Deployments

- **GitHub User Profile:** https://github.com/kartikeysingh-SK
- **Project 01 (Portfolio) Repository:** https://github.com/kartikeysingh-SK/KartikeySingh_PortfolioWebsite
- **Project 01 (Portfolio) Live URL:** https://kartikeysingh-sk.github.io/KartikeySingh_PortfolioWebsite/
- **Project 02 (PTMS) Repository:** https://github.com/kartikeysingh-SK/deepak-tempo-ptms-main
- **Project 02 (PTMS) Live Deployment:** https://deepak-tempo-ptms-main-pzc7.vercel.app/
- **Project 03 (RTRAE) Repository:** https://github.com/kartikeysingh-SK/resource-allocation
- **Project 03 (RTRAE) Live Deployment:** https://resource-allocation-lilac.vercel.app/
- **Academic Submission Upload Drive:** Google Drive Upload Folder

6.3 Conclusion

The **Personal Portfolio Website** and companion academic systems successfully demonstrate comprehensive mastery of core web technologies (HTML5, CSS3, JavaScript) and industry-standard version control methodologies using Git and GitHub. Through systematic planning, modern UI/UX design, accessible component engineering, thorough testing across multiple viewports, and continuous deployment configuration, all criteria stipulated for course **IT4048** have been rigorously fulfilled.

The resulting website and deployed web applications stand as authentic, high-performance evidence representing Kartikey Singh's academic progress at Asha M. Tarsadia Institute of Computer Science and Technology, providing a solid foundation for future technical endeavors.